



- **Project Title: AI-Driven Risk Management and Decision Support for Sustainable Supply Chains**
- **Project Start Date:** 1st June 2025
- **Names of project associates/ students (if student, mentions degree and year):**
 - Puneet, Project Associate
 - Manish Kumawat
- **Problem Statement/ Description (3-4 lines):** Supply chains face unpredictable disruptions that can severely impact service levels and costs. Stochastic risk assessment enables FedEx SMART Centre to quantify these uncertainties using probabilistic models. This approach supports robust decision-making under uncertainty, aligning with FedEx's goal of building resilient, sustainable, and agile logistics solutions.

Progress against project objectives:

S. No	Objective	Progress Status
1.	Risk Assessment Framework	● Working on designing a stochastic risk assessment model integrating probability distributions of supply chain disruptions. ● Found a dataset and worked on the code to implement Monte Carlo simulations to evaluate potential outcomes of supply chain decisions under uncertainty. ● Completed Phase 1 (Literature & Concept Review), including extracting mathematical frameworks, Bayesian Network structures, and ripple-effect modeling from selected papers. ● Developed a comparison matrix of major stochastic modeling approaches. Identified and prepared publicly available datasets, conducted preliminary data analysis (EDA), and flagged data quality issues. ● Completed Monte-Carlo simulation v1 and v2, and drafted the first possible Supply Chain Risk Index (SCRI).

		● Finalized the Supply Chain Risk Index (SCRI) with calibrated severity parameters, achieving $\text{ROC-AUC} \geq 0.80$ and calibration slope within $[0.9, 1.1]$. ● Completed sensitivity and stress testing, along with distribution fitting and copula selection (Gaussian vs. Student-t). ● Incorporated calibration using EM-DAT and UNCTAD external datasets, ensuring mapping accuracy with $R^2 \geq 0.5$. ● Risk Dashboard prototype initiated and integrated with updated SCRI weights, downloadable outputs, VaR/TVaR readouts, and copula model badge. ● With these milestones, the Risk Assessment Framework is now considered functionally complete, providing a strong foundation for the next phase (AI-based decision support system).
2.	AI-Based Decision Support System	● Built a Gym/SimPy environment with calibrated marginals and a Student-t copula sampler, ensuring seeded reproducibility and unit tests for robustness. ● Completed grid-search for (s,S) and myopic baseline policies, and established a Q-learning/DQN training harness with logged KPIs (service level, total cost, SCRI violations, VaR/TVaR). ● Introduced SCRI-penalty and CVaR-style reward terms; retrained agents and validated improvements in tail-risk metrics ($\geq 10\%$ reduction in VaR95 or $\geq 15\%$ fewer SCRI violations without significant cost increase). ● Produced Tech Report v0.5 with environment specification, baselines, QL/DQN results, ablation studies (independence vs Gaussian vs t-copula), and dashboard integration showcasing policy suggestions vs baselines. ● Significant progress was achieved toward deepening the theoretical and mathematical foundations of our reinforcement learning framework. We systematically built mathematical intuitions behind Q-Learning and Deep Q-Networks (DQN), analyzed convergence behavior, and explored integration mechanisms of heavy-tailed dependency models (Student-t Copulas) within the Bellman update structure. In parallel, we reviewed advanced literature on stochastic decision processes and initiated the formulation of a risk-sensitive Bellman operator that embeds copula-based correlation structures

		into value updates. This work lays the groundwork for developing a theoretically consistent and risk-aware AI agent for supply chain decision-making.
3.	Predictive Analytics for Disruptions	Will start exploring this module in the month of November.
4.	Sustainable and Resilient Logistics Strategies	Yet to do

Work completed in the project thus far:

1. Initial Research and Planning: Identified and documented publicly available datasets relevant to supply chain disruption modeling, and researched stochastic risk assessment models (Stochastic Programming, Markov Chains, Bayesian Networks, Newsvendor) with their pros, cons, and application domains.
2. Paper Review and Concept Extraction: Reviewed a key paper on "Stochastic Integrated Supplier Selection and Disruption Risk Assessment under Ripple Effect," summarizing its objective function, decision variables, constraints, uncertainty modeling, Bayesian Network structure, and ripple-effect modeling.
3. Literature Comparison: Created a matrix comparing four major stochastic modeling approaches, highlighting application domains, pros, cons, and references.
4. Seminar Participation and Summaries: Attended the FedEx/IIT Kharagpur seminar on Variance-Adjusted Cosine Similarity (VACS), summarizing key takeaways, and attended the SMART FedEx Seminar Series, summarizing insights on logistics challenges in India, global best practices, and the need for localized solutions.
5. Data Preparation (Phase 2): Reviewed the selected dataset, created a data dictionary mapping original fields to risk variables, identified data quality issues, and conducted exploratory data analysis (EDA), including histograms and ECDFs.
6. Collaboration and Repository Management: Coordinated work with team members, uploaded artifacts to a shared Google Drive and GitHub repository, and maintained logical folder structures with basic documentation.
7. Feature Engineering: Derived inter-arrival time, disruption severity proxy (Delivery_Risk x Cost), and lead-time variance, documented in feature_eng.md.
8. Data Quality Refinement: Proposed imputation/deletion rules per variable and documented them in data_quality_issues.ipynb, with updates to the README file.
9. Exploratory Statistics Polish: Added markdown interpretations for identifying heavy tails and multimodality in exploratory_statistics.ipynb, and pushed histogram and ECDF plots as PNGs to /figures/histogram_ecdfs.
10. Monte Carlo Simulation Development:
 - a. Simulated 10,000 scenarios using preliminary fitted distributions and plotted the distribution of total disruption cost (mc_simulation_v1.ipynb).
 - b. Created Monte-Carlo simulation version 2, including relevant empirical data and a separate parameters folder.
11. Copula Dependency Prototype: Developed a prototype for Gaussian copula dependency on severity and inter-arrival times, documented in copula_dependency_prototype.ipynb.
12. Supply Chain Risk Index (SCRI) Draft: Drafted the first possible SCRI and saved it in the repository as scri_draft.md.

13. Finalized SCRI: Completed the final SCRI, including an iPython notebook and CSV results. Sensitivity and Stress Testing: Performed sensitivity and stress testing analyses.
14. Calibration with External Data: Collected EM-DAT (2018–2024) and UNCTAD port metrics, aggregated to weekly granularity, and mapped disruption severity proxies. Built calibration.ipynb and validated mapping with $R^2 \geq 0.5$.
15. Copula Selection and Diagnostics: Compared Gaussian vs. Student-t copulas on calibrated marginals. Reported AIC/BIC, log-likelihood, Kendall's τ , and tail-dependence λ . Selected the Student-t copula based on fit and tail exceedance accuracy.
16. SCRI Re-weighting and Validation: Re-weighted the Supply Chain Risk Index (SCRI) with calibrated severity parameters. Recomputed node-level scores, achieving ROC-AUC ≥ 0.80 and calibration slope within $[0.9, 1.1]$, using Platt scaling and isotonic regression.
17. Dashboard Enhancements: Updated the Streamlit risk dashboard with model badge (e.g., "t-copula"), downloadable SCRI results, VaR/TVaR readouts, and a short "What changed after calibration" note.
18. MDP Specification & KPIs: Finalized Markov Decision Process (MDP) formulation with defined KPIs including service level, total cost, SCRI-violation count, and VaR/TVaR at weekly horizons. Documented and pushed to GitHub.
19. Simulation Environment Development: Built a Gym/SimPy environment using calibrated marginals with Student-t copula sampling. Added seeded reproducibility and unit tests to ensure deterministic behavior across experiments.
20. Baseline Policy Implementation: Implemented and benchmarked (s,S) and myopic baseline policies. Conducted grid-search runs, logged KPIs, and produced comparative plots and CSV reports for baseline vs. Q-learning performance.
21. Learning Agent Prototyping: Developed initial Q-learning and DQN harnesses, including experience replay, target networks, and logging of learning curves. Saved policies and configs under structured artifact directories for reproducibility.
22. Risk-Aware Reinforcement Learning: Extended agent reward functions with SCRI-penalty and CVaR-style terms. Performed ablation studies (independence vs. Gaussian vs. t-copula; severity $\pm 20\%$, inter-arrival variability $\pm 15\%$). Validated that risk-aware agents improved tail-risk metrics ($\geq 10\%$ reduction in VaR95 or $\geq 15\%$ fewer SCRI-violations) without $> 5\%$ cost escalation.
23. Reinforcement Learning Foundations and Mathematical Deep Dive: Conducted a comprehensive study to strengthen the mathematical underpinnings of Q-Learning and Deep Q-Networks (DQN). Derived the Bellman expectation and optimality equations from first principles, examined convergence conditions under stochastic reward structures, and mapped how parameterized Q-functions approximate the true value function. Documented learning dynamics (exploration–exploitation, discount factor effects, target update stability) and created a unified reference for all reinforcement learning agents developed in the project. This mathematical deep dive has enhanced the theoretical consistency and interpretability of our implemented agents.
24. Copula Integration Research and Student-t Formulation for Risk-Aware Bellman Updates: Explored and mathematically formulated how dependency structures captured by Student-t Copulas can be integrated into the Bellman update equation. Studied joint tail dependencies between cost and disruption variables, derived conditional expectation expressions, and identified potential modifications to the reward and transition components to embed heavy-tailed correlations. This investigation establishes the theoretical pathway toward a copula-augmented, risk-sensitive Q-learning framework, enabling more robust and realistic decision policies under correlated uncertainty.

Specific work done this month:

- Mathematical Exploration of Reinforcement Learning Foundations
 - Built comprehensive mathematical intuition behind Q-Learning and Deep Q-Networks (DQN) by deriving Bellman expectation and optimality equations from first principles.
 - Analyzed the role of value iteration, convergence guarantees, discount factor sensitivity, and the impact of stochastic transition probabilities.
 - Documented detailed derivations and added explanatory markdowns linking theoretical equations with their implementation in the existing RL environment.
 - Conducted collaborative knowledge transfer (KT) sessions for junior members, explaining the QL/DQN learning loop, replay buffer mechanics, and stability techniques.
- Documentation of Q-Learning and DQN Modules
 - Updated and extended the internal documentation in the project repository (project_docs/) covering: Formal definitions of the Bellman operator, temporal-difference update, and loss gradient formulation in DQN. A new section titled “Relevance of QL/DQN for Supply Chain Decision-Making”, establishing the connection between RL agents and FedEx’s real-world logistics decisions under uncertainty.
 - Prepared structured summaries and code-level references that serve as a mathematical and implementation bridge for future model enhancements.
- Study of Student-t Copula and Risk Correlation Modeling
 - Initiated a deep mathematical review of Student-t copula dependency structures to model heavy-tailed joint risk behaviors between disruption frequency, cost, and service reliability.
 - Compared Gaussian and Student-t copulas theoretically through tail dependence λ , Kendall’s τ , and copula density contours, emphasizing their effects on rare event capture.
 - Began formal derivations on how Student-t copula-based dependency can be embedded into the Bellman update equation, introducing correlation terms in the reward-to-value propagation step.
- Collaborative and Reporting Activities
 - Prepared and submitted presentation slides for the FedEx project update session on October 16, 2025, highlighting theoretical contributions and planned integrations.
 - Drafted and finalized the dataset request email to FedEx, ensuring alignment with model validation needs and the confidentiality clauses of the NDA.
 - Updated the GitHub project repository with refined documentation (rl_docs/, math_notes/, copula_derivations.md) and synchronized progress with the advisor’s feedback.
 - Maintained weekly communication with the project mentor and provided detailed updates summarizing conceptual and mathematical advancements.
- Integration Pathway Design for Risk-Aware Bellman Operator
 - Conducted symbolic experiments (using Python sympy) to validate mathematical consistency and identify feasible points of integration in the existing DQN architecture.

Project Outcome for the month:

- Strengthened the theoretical foundation of the AI-based decision support system by mathematically deriving and validating Q-Learning and DQN update formulations, ensuring conceptual integrity in the agent design.
- Developed a comprehensive mathematical framework for integrating Student-t copula-based dependency structures into the Bellman update, establishing a robust pathway toward risk-aware reinforcement learning.
- Produced in-depth documentation and derivation reports, bridging theoretical insights with practical implementation notes to aid future reproducibility and model transparency.
- Enhanced the project repository with updated QL/DQN documentation, mathematical appendices, and copula integration notes, consolidating all research under a unified and auditable structure.
- Advance the transition from theoretical exploration to implementation readiness, ensuring that next month's objectives (risk-sensitive Bellman implementation and simulation experiments) rest on a rigorous analytical foundation.
- Maintained consistent communication and feedback with the project mentor, ensuring alignment of academic rigor with FedEx's applied objectives in stochastic decision-making and resilience modeling.

Next Steps (Work Plan for next month):

- Begin active exploration of the Predictive Analytics module, focusing on identifying key disruption drivers and forecasting techniques relevant to FedEx operations.
- Design preliminary feature sets and model structures to support disruption prediction and reliability forecasting once operational data is received.
- Upon receipt of the dataset, initiate data validation, preprocessing, and exploratory analytics to align the data schema with existing risk and reinforcement learning modules.
- Formally derive and test the final equation update for incorporating the Student-t copula dependency structure into the Bellman update rule of Q-Learning and DQN frameworks.
- Conduct numerical simulations to empirically verify the impact of the copula-augmented Bellman formulation on stability, convergence, and tail-risk reduction.
- Document mathematical findings, simulation outcomes, and visual results to support integration into the next technical report and dashboard release.

Project Progress:**Project Timeline and current work activity:**

Phase/Work Tasks	Sub-Tasks	June	July	August	September	October
3.1 Risk Assessment Framework	Literature review and model design	Completed				

	Paper extraction & modeling approach	Completed				
	Data sourcing and initial mapping	Completed				
	Data cleaning and quality assessment		Completed			
	Exploratory data analysis and distribution fitting		Completed			
	Monte Carlo simulation prototype		Completed			
	Multi-factor risk index design			Completed		
	Risk index testing and refinement		Completed			
	Distribution Fitting & Goodness-of-Fit (GoF) Matrix		Completed			
	Alternative Copula Check			Completed		
	Sensitivity and Stress Testing			Completed		
	Risk Dashboard Prototype			Completed		
	Deployment and DevOps			Completed		
AI-Based Decision Support System	Environment and Baseline Setup				Completed	
	Baseline Implementation and Training Harness				Completed	
	Risk Aware Learning and Analysis				Completed	
	Deep Dive into RL Agents					Start
	Mathematical Study of Student t Copula					Start
	FedEx terminology research for dataset preparation					Start

	Packaging and Demo Preparation					Start
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Work Contributions by individual team members in the project:

1. Puneet

- a. Led the theoretical exploration of reinforcement learning foundations, deriving and documenting the mathematical basis of Q-Learning and DQN from first principles, including Bellman expectation and optimality equations.
- b. Created a detailed internal reference note connecting mathematical equations to implementation, covering convergence behavior, exploration–exploitation trade-offs, and update stability.
- c. Authored complete documentation for both Q-Learning and DQN agents, integrating theoretical intuitions with project-specific design explanations in the project_docs/ folder.
- d. Conducted an in-depth study of Student-t copula models to understand heavy-tailed dependencies and their effect on risk propagation through the Bellman update rule.
- e. Designed a copula-augmented Bellman formulation, exploring how Student-t dependencies could be embedded into Q-value updates for risk-aware decision-making.
- f. Drafted analytical notes on “Tail-Risk Propagation and Copula-Augmented Value Functions,” setting the foundation for next month’s implementation experiments.
- g. Managed all repository updates, documentation, and mathematical reports, including new markdowns (math_notes/, copula_derivations.md) and reinforcement learning references.
- h. Coordinated communication with the project mentor, prepared and presented the October progress presentation, and finalized the dataset request draft for FedEx validation data.
- i. Ensured that all findings, derivations, and references were versioned, reproducible, and traceable in the GitHub repository for ongoing research continuity.